QUANTITATIVE ANALYSIS

What is Titration?

To determine the concentration of an unknown sample, we use a common laboratory technique called volumetric analysis. In this quantitative technique, the concentration of the particular substance present in an unknown sample solution can be determined by measuring the volume of another solution that reacts with it. The solution that has been taken in the burette is called the titrant and the solution taken in the conical flask is called the analyte.

What is end point of a titration?

When the reaction between fixed volume of the titrant and the analyte becomes complete, the titration is said to be reached the end point. This end point can be identified using suitable indicators.

What are indicators?

An indicator is a chemical substance that undergoes a sharp colour change at the end point. In acid-base titrations, the end point is usually determined using acid-base indicators or pH indicators. Acid Base indicators are either weak organic acids or weak organic bases.

What is a standard solution?

A solution whose concentration is known is called a standard solution. It can be prepared by dissolving a known quantity of the substance in a definite volume of the solvent.

How do we determine the strength of a given acid or base?

Determination of the strength is based on the Law of Equivalents. According to this law, the number of equivalence of the substance to be titrated is equal to the number of equivalence of the titrant used.

Consider an acid-base titration in which V_1 ml of an acid solution of normality N_1 is required to neutralize V_2 ml of a base of normality N_2, then according to normality equation

$$V_1N_1=V_2N_2$$

From the above equation the normality of unknown solution can be calculated by measuring the volume of standard solution consumed.

Apparatus Used in Volumetric Analysis

The following are the apparatus required in volumetric analysis.

(i) Graduated-burette.
(ii) 20 ml pipette
(iii) Conical flask.
(iv) Funnel, weighing machine, wash bottle, etc.,.

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